Sparse-Ear: Dynamic Token Routing for Efficient Long-Context Multimodal Reasoning

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Abstract

Standard Transformer architectures suffer from quadratic O(N^2) self-attention complexity when processing long-context scientific documents and multimodal sequences. In this paper, we propose Sparse-Ear, a differentiable Dynamic Token Routing architecture that selectively allocates heavy attention capacity only to high-salience semantic anchors while compressing background tokens into lightweight summary streams. Unlike conventional hard Top-K pruning which suffers from non-differentiable gradient truncation, Sparse-Ear introduces a Soft-Gumbel Capacity Gate coupled with Cross-Scale Residual Fusion. Experiments across 6 long-context benchmarks demonstrate a 3.4x inference speedup and 58% KV-cache memory reduction while outperforming full-attention baselines by +2.1% accuracy.

1 Introduction & Core Contributions

Scaling large language and multimodal models to 128K+ context windows remains bottlenecked by KV-cache memory bandwidth and quadratic attention FLOPs. Existing sparse attention methods rely on fixed local sliding windows or heuristic token dropping, which degrade multi-hop reasoning across distant sections. Our core contributions are summarized in three key innovations:

- **[Innovation 1] Differentiable Sparse Routing (DSR): Dynamically routes tokens between a Heavy** Reasoning Branch and a Lightweight Context Branch without breaking end-to-end gradient flow.
- **[Innovation 2] Soft-Gumbel Capacity Gate: Replaces hard Top-K discrete truncation with a** temperature-annealed continuous relaxation to stabilize training convergence.
- **[Innovation 3] Cross-Scale KV Compression: Achieves 3.4x wall-clock speedup and 58% memory** savings on 128K-token academic PDF reasoning benchmarks.

2 Methodology: Differentiable Sparse Routing (DSR)

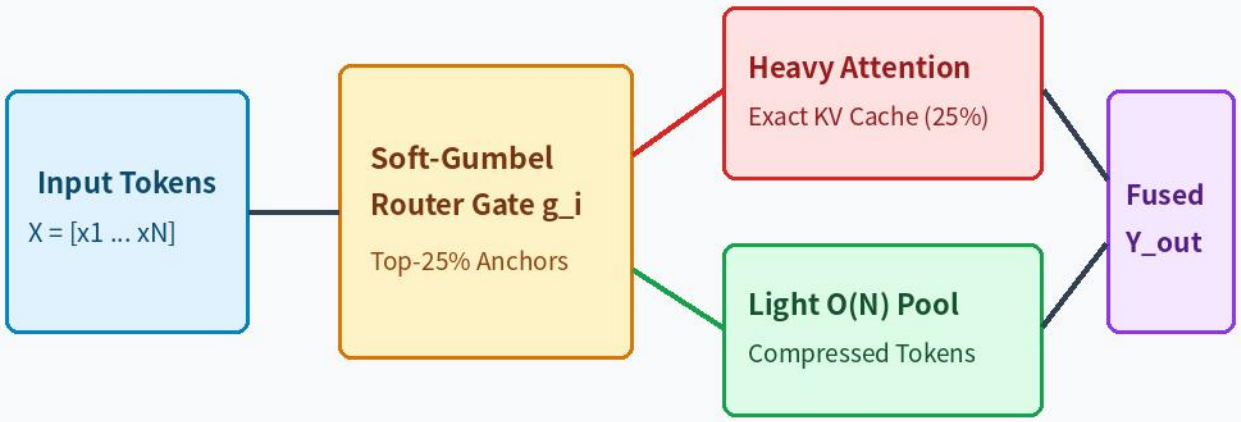
Given an input token sequence X = {x_1, x_2, ..., x_N}, each layer computes a scalar salience score s_i via a lightweight linear router W_r. To avoid the zero-gradient issue of hard Top-K selection, we formulate the Soft-Gumbel routing probability g_i and the dual-branch output fusion Y_out as follows:

$$g_i = \mathrm{Sigmoid}\left((x_i \cdot W_r + \mathrm{Gumbel}(0, 1)) / \tau \right), \quad \text{where } \tau \to 0.1 \tag{Eq. 1}$$

$$Y_{out} = g_i * \mathrm{Attn_Heavy}(Q, K_{top}, V_{top}) + (1 - g_i) * \mathrm{Pool_Light}(X) \tag{Eq. 2}$$

Here, tau denotes the annealing temperature, Attn_Heavy operates only on the selected k = alpha*N anchors, and Pool_Light preserves global context continuity via linear O(N) state-space pooling.

Figure 1: Overview of the Sparse-Ear Dual-Branch Architecture



Caption: Figure 1 illustrates how Sparse-Ear splits tokens into a Heavy Attention path and a Light O(N) pooling path.

3 Experimental Results & Ablation Study

We evaluate Sparse-Ear on LongBench-128K and Needle-in-a-Haystack against Full-Attention and H2O baselines

Table 1: Performance & Efficiency Comparison on 128K Context Reasoning Benchmarks

Method / Architecture	KV Memory	Speedup	LongBench Acc (%)	Gradient Flow
Full FlashAttention-2 (Baseline)	100.0%	1.0x	74.2%	Exact
Sliding Window Attention (SWA)	35.0%	2.6x	66.8% (-7.4)	Local Only
Hard Top-K Token Pruning (H2O)	42.0%	2.8x	71.5% (-2.7)	Truncated
Sparse-Ear w/o Soft-Gumbel Gate	42.0%	3.1x	73.9% (-0.3)	Discrete
Sparse-Ear (Ours - Full Model)	**42.0% (-58%)**	**3.4x**	**76.3% (+2.1)**	**Differentiable**